

DURHAM 11 W, NC, USA

WMO: 756940

Lat: 35.9705N

Lon: 79.0931W

Elev: 171

StdP: 99.28

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 03758

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-5.6	-18.3	0.8	-5.8	-15.2	1.0	-2.7	4.8	10.8	4.4	8.2	1.1	N/A	0.341

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	33.8	23.3	32.5	23.2	31.4	22.9	25.2	30.7	24.5	29.9	24.0	29.2	2.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.6	18.8	27.5	23.0	18.1	26.9	22.4	17.5	26.3	77.5	30.8	74.9	30.0	72.7	29.3	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
4.2	3.7	3.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-11.9	35.5	2.9	1.2	-14.0	36.4	-15.8	37.1	-17.4	37.8	-19.5	38.7
			-12.4	26.1	2.5	0.8	-14.2	26.6	-15.6	27.1	-17.1	27.5	-18.9	28.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	4.0	6.5	10.1	15.2	19.8	23.7	25.3	24.7	21.9	15.8	9.4	6.9
	DBStd	8.59	5.56	5.64	5.44	4.50	3.71	2.61	2.21	2.33	3.14	4.47	4.50	5.22
	HDD10.0	603	198	127	69	9	0	0	0	0	0	7	64	129
	HDD18.3	1921	446	333	261	112	31	1	0	0	8	103	269	357
	CDD10.0	2546	11	29	71	166	303	412	476	456	358	187	46	32
	CDD18.3	822	0	1	4	19	76	163	217	198	116	25	1	1
	CDH23.3	6922	0	6	34	197	591	1469	1926	1642	856	196	4	1
	CDH26.7	2587	0	0	2	34	159	566	816	660	307	44	0	0
(14) Wind	WSAvg	1.4	1.5	1.7	1.7	1.7	1.4	1.2	1.2	1.1	1.2	1.2	1.3	1.4
(15) Precipitation	PrecAvg	1186	101	93	111	84	104	105	108	116	99	90	86	83
	PrecMax	1556	227	172	202	187	194	258	214	230	610	257	319	182
	PrecMin	846	30	14	39	18	28	22	7	20	2	7	14	8
	PrecStd	181	49	40	47	41	42	57	51	48	100	59	60	39
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.6	24.1	26.6	29.6	32.0	34.2	35.5	35.5	33.6	30.4	23.5	22.7
		MCWB	14.9	17.0	15.7	17.9	21.8	23.7	23.6	22.8	21.6	21.7	16.8	18.1
	2%	DB	18.1	21.3	24.1	27.6	30.0	32.7	33.8	33.3	31.6	27.6	21.5	20.3
		MCWB	13.2	14.6	15.1	17.7	21.0	23.3	23.7	23.3	21.3	20.3	16.0	16.9
	5%	DB	15.7	18.9	22.0	25.7	28.5	31.5	32.5	31.7	30.1	25.4	19.7	18.1
		MCWB	11.4	13.9	13.8	17.0	20.4	23.1	23.8	23.2	21.3	18.6	14.9	15.6
	10%	DB	13.1	16.5	19.3	23.7	26.9	30.1	31.0	30.4	28.5	23.6	17.5	16.0
		MCWB	9.3	12.1	12.9	16.1	19.8	22.5	23.4	22.7	20.9	18.1	13.1	12.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.2	17.9	18.0	21.4	23.4	25.2	26.3	25.8	24.2	23.5	19.3	19.4
		MCDB	19.5	21.8	23.0	25.4	28.4	31.4	32.0	30.9	28.5	26.5	21.3	21.2
	2%	WB	15.1	16.4	16.9	19.8	22.2	24.4	25.3	25.0	23.4	22.1	17.7	17.6
		MCDB	17.4	19.9	21.6	24.4	27.6	30.5	31.1	30.0	28.0	25.3	20.3	19.3
	5%	WB	11.9	14.9	15.8	18.6	21.5	23.8	24.7	24.2	22.7	20.6	16.2	15.8
		MCDB	13.9	18.3	20.0	22.9	26.8	29.7	30.4	28.8	27.2	24.3	18.8	17.8
	10%	WB	9.8	12.3	14.6	17.5	20.8	23.1	24.1	23.7	21.9	18.8	14.4	13.0
		MCDB	13.2	15.8	18.4	22.1	25.8	28.7	29.8	28.0	25.9	22.4	17.3	15.3
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	11.3	11.8	12.5	11.1	11.2	10.7	10.4	10.5	11.8	11.9	10.2
		MCDBR	14.0	13.7	13.8	14.5	12.7	12.4	12.2	12.3	13.1	13.3	13.8	11.4
		MCWBR	9.9	8.8	7.2	6.3	5.4	4.8	4.2	4.1	4.8	6.6	8.5	8.3
	5% WB	MCDBR	11.3	12.3	12.0	11.0	10.7	11.4	10.8	9.8	10.2	10.5	10.3	9.8
		MCWBR	10.0	9.2	7.2	6.0	4.8	5.0	4.3	4.0	4.5	5.9	8.0	8.1
(40) Clear-Sky Solar Irradiance	taub		0.311	0.328	0.361	0.402	0.449	0.495	0.536	0.526	0.420	0.379	0.329	0.314
	taud		2.504	2.447	2.394	2.280	2.182	2.095	1.997	2.010	2.304	2.396	2.497	2.528
	Ebn at Noon		890	911	907	882	842	800	764	762	835	842	860	864
	Edh at Noon		82	97	112	132	148	161	176	171	121	101	82	75
(44) All-Sky Solar Radiation	RadAvg		2.55	3.23	4.27	5.37	5.94	6.38	5.99	5.44	4.53	3.68	2.80	2.20
	RadStd		0.17	0.39	0.30	0.40	0.47	0.45	0.37	0.28	0.40	0.47	0.22	0.19

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+146	+99

Nomenclature: See separate page